

Customer Churn Prediction

September 12, 2026

Turning customer signals into proactive retention.



From raw behavior to retention action

PURPOSE

Identify customers likely to churn before they leave.

01 Analyze customer patterns

02 Engineer predictive features

03 Develop & validate models

04 Translate outputs into action

Retention is the cheaper growth engine.

15-25%

average annual industry churn

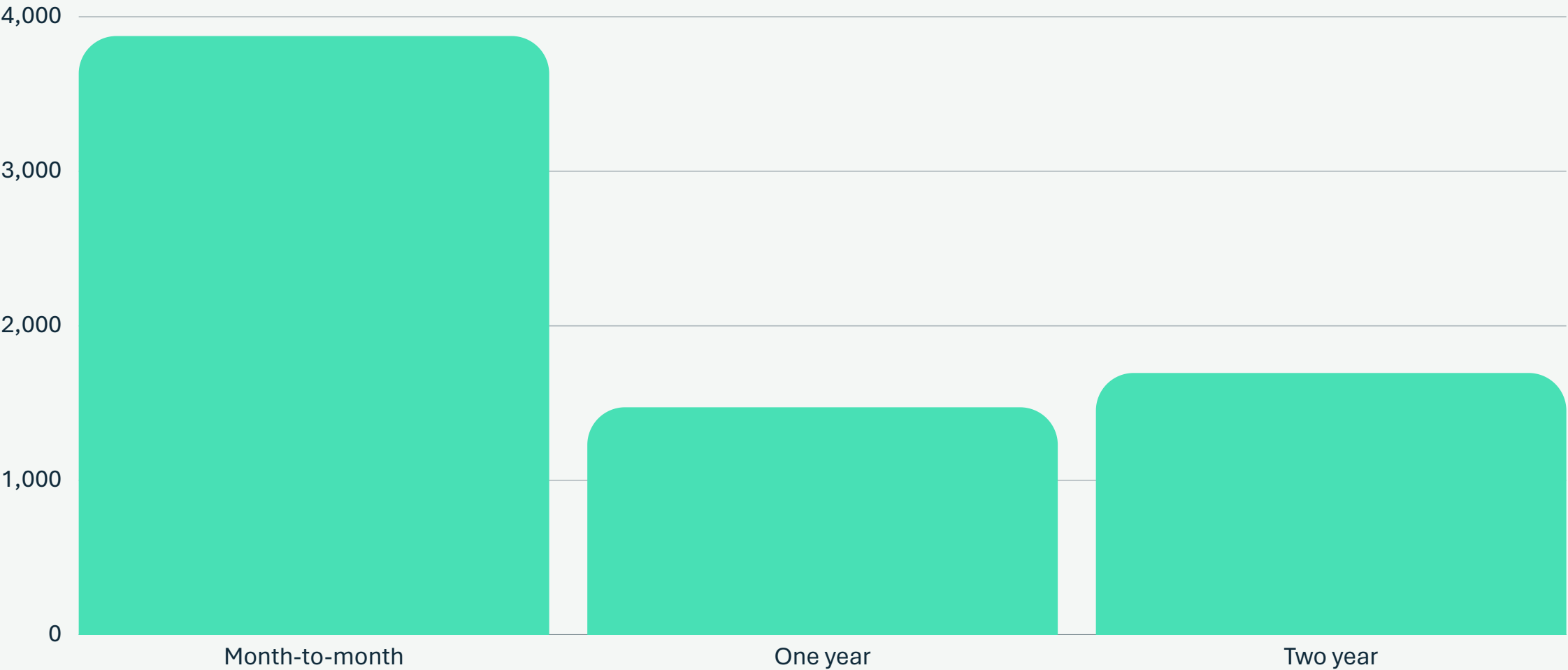
5×

more expensive to acquire a new customer than
retain an existing one

-7,000 customer records

Demographics • contract type • service usage
payment method • tenure • monthly charges

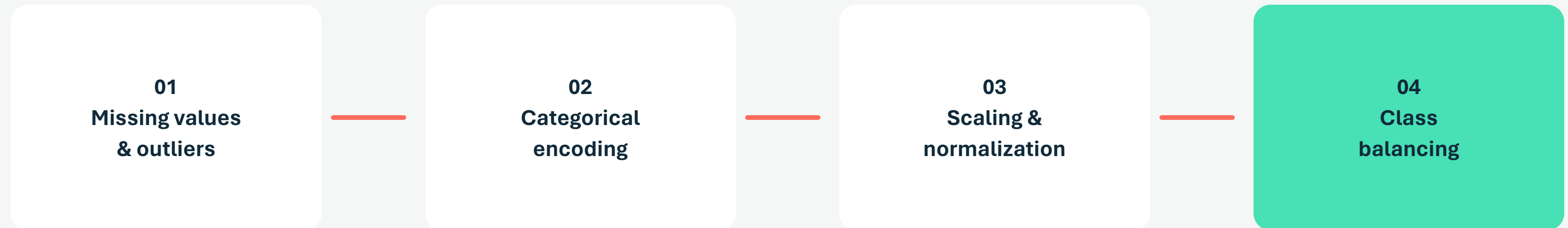
Customer distribution by contract type



PUBLIC DATASET

Telco Customer Churn

A clean pipeline prevents noisy predictions

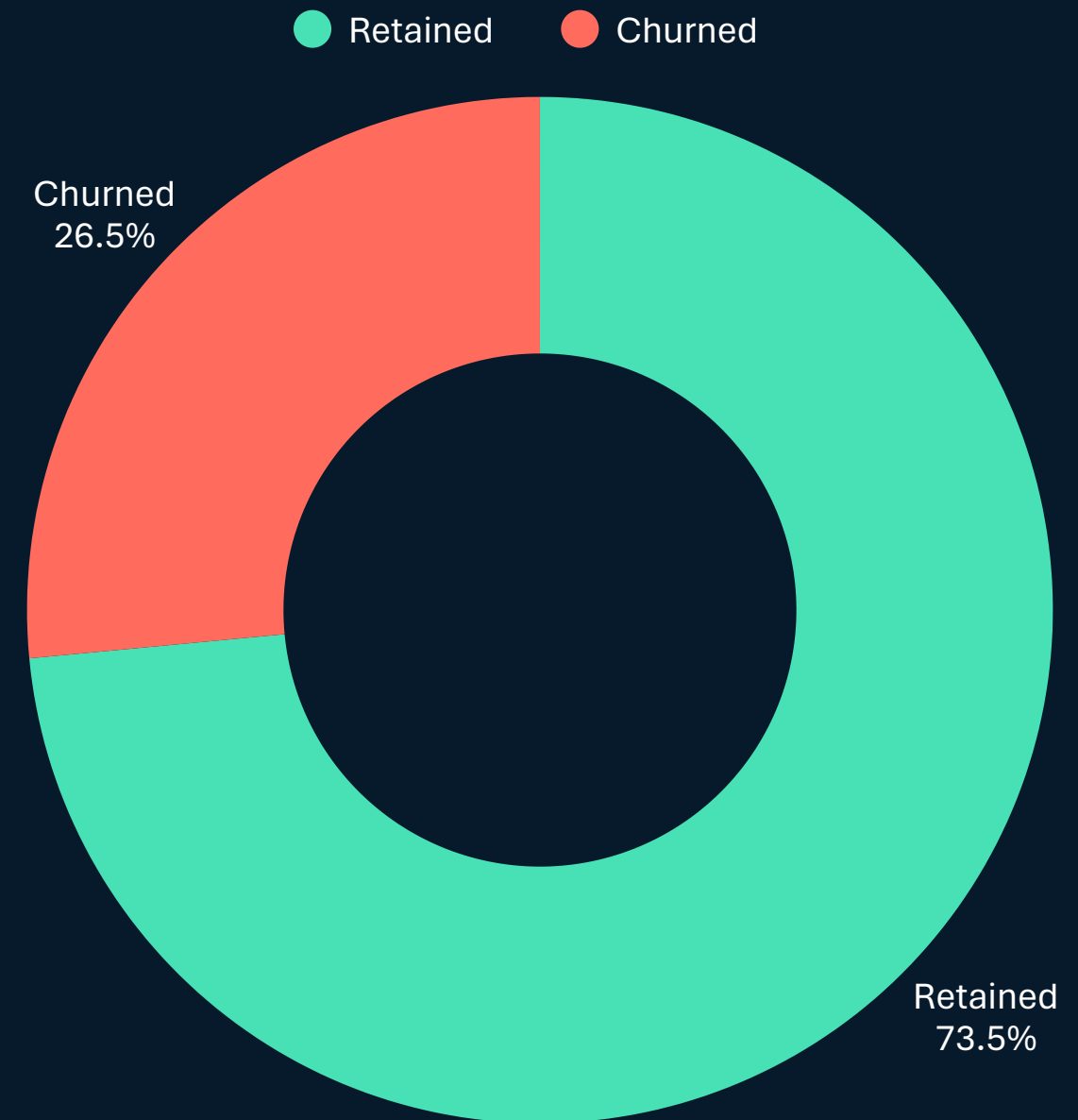


Oversampling and undersampling were tested to improve minority-class recall.

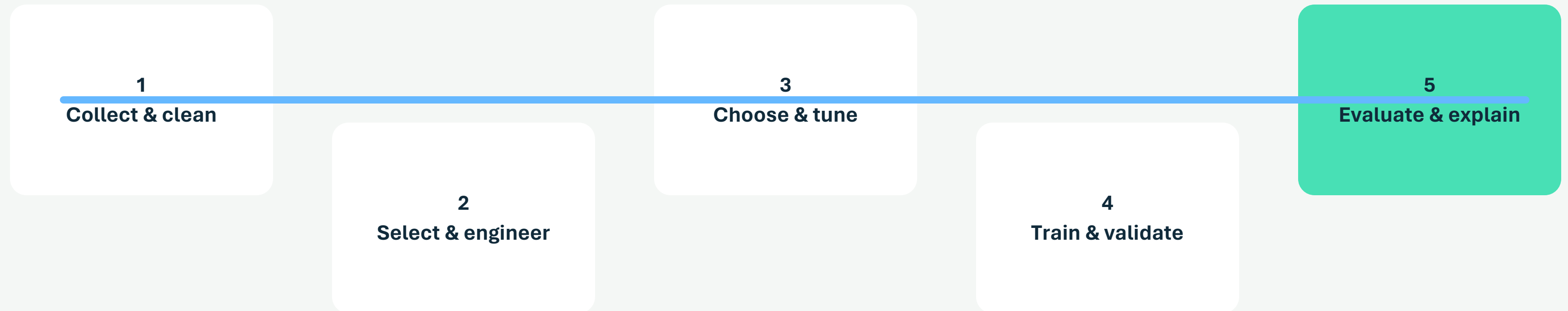
Churn concentrates where commitment is lowest.

High monthly charges + short tenure
create the strongest churn signal.

Customer outcome distribution



Five phases. One reliable decision system.



IMPLEMENTATION

Built in Python. Designed to iterate.

Jupyter Notebook

pandas • scikit-learn

matplotlib • imbalanced-learn

TELCO CUSTOMER CHURN / PUBLIC DATASET



Four models entered. XGBoost led the balance.

1 Logistic Regression

2 Decision Tree

3 Random Forest

4 Gradient Boosting (XGBoost)

FINAL MODEL

XGBoost

Best accuracy + recall balance

Performance, viewed from every angle

Accuracy

Overall correctness

Precision

Quality of churn alerts

Recall

Coverage of actual churn

F1-Score

Precision–recall balance

Confusion Matrix

TP • FP • TN • FN

RESULTS & KEY FINDINGS

87% accuracy— with useful recall.

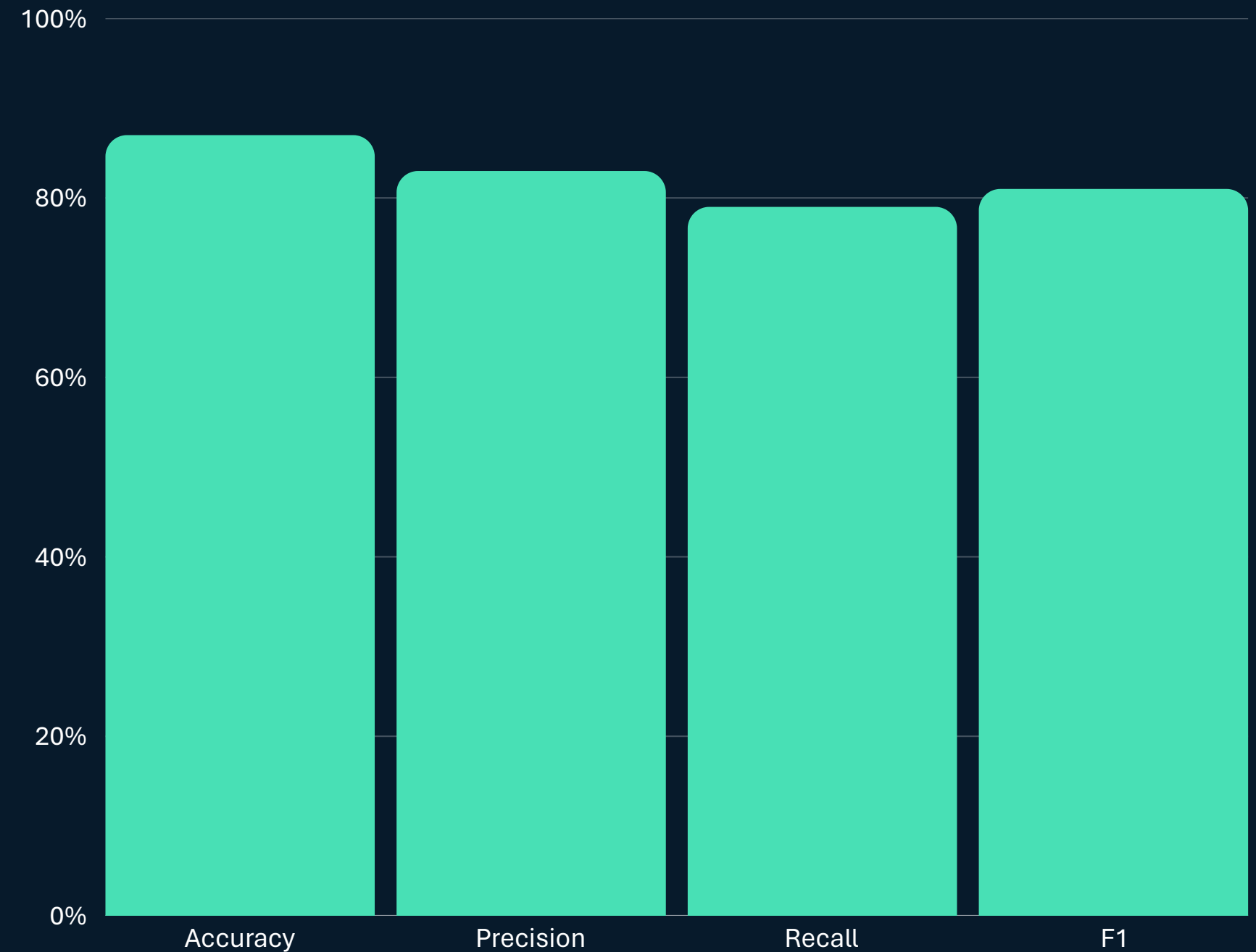
Top predictors

Tenure • Monthly Charges • Contract Type

+15%

recall after preprocessing
and class balancing

XGBoost performance



What the project taught—and what comes next

CHALLENGES

Imbalanced classes
Hyperparameter tuning
Model interpretation

LEARNINGS

EDA before modeling
Preprocessing discipline
Explainability matters

FUTURE SCOPE

Production deployment
Customer feedback loops
Behavioral data

LIMITATIONS / Dataset size • no real-time data • incomplete behavioral features

CONCLUSION

Predict earlier. Retain smarter.

A robust churn model enables proactive, targeted retention—reducing attrition while protecting customer lifetime value.

Thank you.

Questions and feedback welcome